

ELECTRONIC DICE USING DOT-MATRIX DISPLAY



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The roll of a dice has decided the fate of kingdoms. The dice is the oldest device known to human beings for generating random numbers from 1 to 6.

Here we present an electronic device using an 8x8 dot-matrix LED display to simulate the faces of a real dice. Pressing a switch (S2 in Fig. 1) generates a random number on the display. A microcontroller is used to check the status of the switch and generate a random number. The dice number is displayed on the dot-matrix LED display with the help of an LED display driver.

Circuit diagram

Fig. 1 shows the circuit of the microcontroller-based dice using dot-matrix display. It comprises an AVR micro-

controller ATmega8 (IC1), column-cathode display driver MAX7219 (IC2), regulator 7805 (IC3), dot-matrix LED display and a few discrete components.

ATmega8 is a low-power, 8-bit microcontroller based on the AVR RISC. It has 8Kb in-system programmable flash memory with read-while-write capabilities, 512 bytes of EEPROM, 1Kb serial random-access memory (RAM), 23 general-purpose input/output (I/O) lines, 32 general-purpose working registers, three flexible timers/counters with compare modes, internal and external interrupts, and a byte-oriented two-wire serial interface.

Port B of the microcontroller is configured as output and used for serial communication with MAX7219 driver, which is a programmable device that controls the information display on dot-matrix LED display DIS1. Port pins

PB0, PB1 and PB2 of microcontroller ATmega8 are interfaced to pin 1 (D_{IN}), pin 13 (CLK) and pin 12 (LOAD) of display driver MAX7219, respectively.

Random number is displayed by means of lighted LEDs. The data and control information is transferred to MAX7219 in 16 clock cycles.

Port pin PD0 of ATmega8 is configured as input and used for checking the status of the play switch (S2). As soon as pressing of the switch is sensed, the microcontroller generates a number to show on dot-matrix display

DIS1. Key de-bounce is implemented with the help of the software.

Display driver MAX7219 interfaces the microcontroller to the LED display. It can control 64 individual LEDs and consists of an on-chip binary-



Fig. 2: Pin configuration of 7805

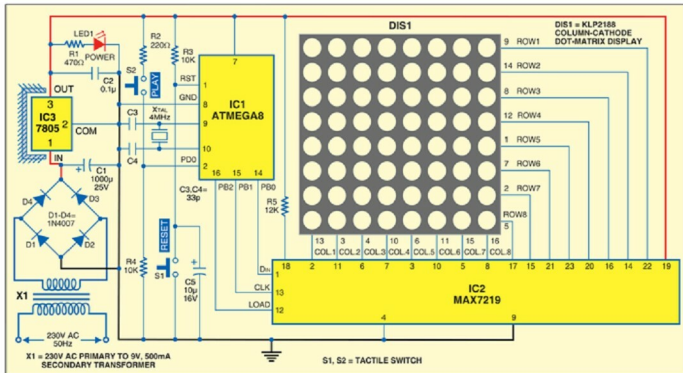


Fig. 1: Circuit of microcontroller-based dice using dot-matrix display